

Akashdeep Ghosh

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PROFILE

Software Engineer with 3+ years building production-grade backend and AI systems. Specialized in designing high-performance data pipelines and LLM-integrated tools that translate complex data into actionable insights. Proven track record of end-to-end ownership - from architecture to production optimization - delivering measurable improvements in system performance and user productivity.

EXPERIENCE

Cisco 01/2026 – Present | Bengaluru, India

Software Engineer

- Contributed to the topology platform, owning the query API layer and persistence path - including unique UAI ID generation, upserts, snapshots, group member resolution, and response transformations - as part of a small team building the entire platform across all layers.
- Designed topology lifecycle management including soft delete, staged deletion, and automated stale-record cleanup, along with health mapping and L3 topology derivation, and automated cleanup of **millions of** stale records per day across the topology platform.
- Architected **Kafka**-driven webhook registration and lifecycle management for **SDWAN** and **Meraki**, including auto-registration, credential lifecycle handling, reconciliation, retries, and idempotent deletion, with tenant and product-tenant-aware scoping - supporting **1M webhook registrations** in production.
- Built Meraki integration layer covering push API topology processing, linked org resolution, scoped registration, and machine-account fallback for 401/403/429 responses, with per-run response caching that cut redundant API calls by **60%**.
- Built SDWAN integration layer with paginated tenant discovery, post-ingestion topology triggering, batch consumption, and negative caching with rate-limit handling, reducing API call volume and failed requests.
- Implemented **MCP** routing for the UAI query service, including auth, tenant context propagation, and skill manifest generation for topology and AI canvas.
- Drove reliability and observability across repos — standardized metrics and outbound API instrumentation, alert runbooks, metric cardinality and auth label hardening, **Kafka** delivery + DLQ classification.

Dassana (Acquired by Deepwatch)

Software Engineer II 02/2025 – 01/2026 | Bengaluru, India

- **Leading** development of an internal **AI-powered chatbot** that enables non-technical stakeholders to query security data using natural language, automatically generating SQL queries and scoped interactive visual insights, accelerating data discovery and reducing client analysis time by **~60%**.
- Designed the full pipeline including safety validation, ability to choose models, text-to-SQL generation, query execution, visualization, and conversational follow-up enabling **seamless multi-turn interactions**.
- Migrated internal package management across 3 major systems (public PyPI → GitHub Packages → AWS CodeArtifact) to comply with new **security and compliance mandates** without any downtime across **20+ Python services**.
- Continued ownership of core ingestion and data systems, while adapting to new infra and workflows post-acquisition, ensuring zero production regressions.

Software Engineer 08/2023 – 02/2025 | Bengaluru, India

- **Optimized ETL pipelines** processing **10–20M+ vulnerabilities** per ingestion cycle for sources like Wiz, Tenable etc. cutting overall pipeline runtime by **~40%**.
- **Developed a standardized ingestion framework** supporting **15+ data sources**, reducing per-source setup time from **1 day to under 1 hour**.
- **Designed and developed event-driven Argo Workflows** orchestrating multiple ingestion CronJobs segmented by type (assets/vulnerabilities) and source, implementing parallel sub-type processing for high-volume sources to eliminate historical bottlenecks.
- **Engineered** Snowflake procedures for stage-to-table merging and data reconciliation; later **migrated normalization logic** from Snowflake → Python → Go reducing processing time by **~50%** by decoupling compute from warehouse constraints and moving compute-intensive operations to scalable infrastructure which reduced compute cost by **~30%**.
- **Revamped** the company-wide monitoring and alerting system (originally Slack-based) into an **event-driven observability stack** to multiple destinations using **Argo, Grafana, and NATS**, improving incident visibility of job failures, pod restarts, uptime and usage metrics and reducing mean time to detect by **~70%**.
- **Collaborated directly** with enterprise customers to diagnose ingestion issues and resolve data quality problems in production environments.

Software Engineer Intern

08/2022 – 08/2023 | Bengaluru, India

- Built a credit-score style security team scoring system **penalizing SLA breaches with exponential score decay** - tracking all teams of an organization, providing CISOs data-driven insights to reallocate resources via competitive leaderboards and which helped improving critical vulnerability remediation compliance through competitive leaderboards.
- **Implemented a Slack-based monitoring and notification system** that tracked ingestion job failures.

Technology Used: Golang, Python, Java, Quarkus, Argo Workflows, NATS, Snowflake, Langchain4j, Langfuse, N8N, Grafana, AWS SES, Docker

EDUCATION

Lovely Professional University

2019 – 2023 | Jalandhar, India

Bachelor of Technology in Computer Science and Engineering

SKILLS

Languages: Python | Java | Golang | SQL

Database & Storage: PostgreSQL | Snowflake | OpenSearch | Valkey

Cloud & Infra: AWS | GCP | Kubernetes | REST API | Docker | Terraform | Argo Workflows

Frameworks: Quarkus | Micronaut | SpringBoot

DevOps & Tools: Git | GitHub Actions | NATS | Grafana | LaunchDarkly